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LETTER

A newly discovered role of evolution in previously published consumer–resource dynamics

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Abstract

Consumer–resource interactions are fundamental components of ecological communities. Classic features of consumer–resource models are that temporal dynamics are often cyclic, with a ¼-period lag between resource and consumer population peaks. However, there are few published empirical examples of this pattern. Here, we show that many published examples of consumer–resource cycling show instead patterns indicating eco-evolutionary dynamics. When prey evolve along a trade-off between defence and competitive ability, two-species consumer–resource cycles become longer and antiphase (half-period lag, so consumer maxima coincide with minima of the resource species). Using stringent criteria, we identified 21 two-species consumer–resource time series, published between 1934 and 1997, suitable to investigate for eco-evolutionary dynamics. We developed a statistical method to probe for a transition from classic to eco-evolutionary cycles, and find evidence for eco-evolutionary type cycles in about half of the studies. We show that rapid prey evolution is the most likely explanation for the observed patterns.

Keywords

Eco-evolutionary dynamics, host–parasite interaction, population cycles, predator–prey interaction, rapid evolution, retrospective analysis.

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INTRODUCTION

Consumer–resource interactions are an essential component of all ecological communities, because all organisms take part as victim, exploiter, or both. As a result, these interactions have received considerable attention from both theoreticians and empiricists (Murdoch *et al*. 2003). The two-species population-level outcome when the members of one taxon kill and consume the other depends upon both the behaviours of the organisms (e.g., search, capture, ingestion, assimilation efficiencies; Stephens & Krebs 1986), and the nature of the environment where they reside (e.g., external forcing or spatial structure; Chesson 2003; Simonis 2012). The resulting temporal predator–prey dynamics, even in simple systems, range from extinction to stable coexistence, to stable oscillations and complex dynamics (Lotka 1925; Volterra 1926; Nicholson & Bailey 1935; Rosenzweig & MacArthur 1963; May 1972; Anderson & May 1981; Kendall *et al*. 1998). Within this diversity of possibilities, stable predator–prey oscillations are a classic two-species pattern in ecology, covered in virtually every introductory textbook whether or not they are regularly observed either in nature or the laboratory (Ives 1998).

Two important properties of consumer–resource cycles are period, the time between successive peaks in predator (or prey) abundance, and phase lag, the time between successive peaks in prey and predator abundance. Theory suggests that the phase lag should typically approximate ¼ of the cycle period (Bulmer 1976). Among the best-known laboratory examples of this ¼-period lag are Huffaker's experiments on mites on oranges (Huffaker 1958) and Luckinbill's ciliate microcosm experiments (Luckinbill 1973, 1974). Phase lags in nature have been reviewed by Bulmer using data on fur-bearing mammals in Canada (Bulmer 1974, 1975). For example, he showed that in the Lynx-Hare time series with a period of 10 years, the peak lynx abundance followed peak hare abundance with a 2-year lag (prediction of ¼ = 2.5 years). But despite these encouraging results, many well-known examples of cyclic consumer–resource dynamics in experimental settings fail even by simple inspection to show a clear ¼-period lag (Gause 1934; Utida 1957; Jost *et al*. 1973; van den Ende 1973; Bohannan & Lenski 1997; Yoshida *et al*. 2003, 2007; see Table 1). Indeed, strong evidence for the typical consumer–resource cycles expected from theory is generally lacking (Ives 1998). Furthermore, what is classified as 'classic' ¼-period lag may depend upon the eye of the beholder: Hutchinson (1978) called Utida's (1957) experimental parasitoid–host cycles 'classic', whereas Hassell (1978) called them 'variable', presumably seeing only parts of the series as classic.

Many published data sets showing consumer–resource oscillations appear to have antiphase or nearly antiphase cycles, rather than a ¼-period lag: i.e. peaks in consumer

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Table 1 Consumer–resource interactions included in this study: interaction type (PP: predator–prey; PH: parasite-host), organisms used and Evolutionary Dynamics Index (EDI; large negative values show transition from quarter-period lag to antiphase cycles, except for data set 14 which became more in-phase rather than more antiphase). Table S2 gives additional information about each data set

		Taxa			
	Interaction type	Consumer	Resource	EDI	Data source
1	PH	Phage	Cyanobacteria	-1^{ns}	Barnet *et al.* (1981)
2	PH	Phage	Bacteria	−91**	Bohannan & Lenski (1997)
3	PH	Bacteria	Bacteria	19^{ns}	Dulos & Marchand (1984)
4	PP	Ciliate	Bacteria	−48**	Canale *et al.* (1973)
5	PP	Ciliate	Bacteria	−39**	Jost *et al.* (1973)
6	PP	Ciliate	Bacteria	−36**	van den Ende (1973)
7	PP	Amoebae	Bacteria	−55**	Tsuchiya *et al.* (1972)
8	PP	Amoebae	Bacteria	−74**	Tsuchiya *et al.* (1972)
9	PP	Amoebae	Bacteria	−47**	Tsuchiya *et al.* (1972)
10	PP	Ciliate	Ciliate	−15**	Gause (1934)
11	PP	Ciliate	Ciliate	13^{ns}	Luckinbill (1973)
12	PP	Ciliate	Ciliate	-2^{ns}	Luckinbill (1974)
13	PP	Ciliate	Ciliate	15^{ns}	Veilleux (1976)
14	PP	Ciliate	Ciliate	−15**	Veilleux (1976)
15	PP	Ciliate	Ciliate	8^{ns}	Veilleux (1976)
16	PP	Rotifer	Algae	−11*	Boraas (1980)
17	PP	Rotifer	Algae	−37**	Yoshida *et al.* (2003)
18	PP	Rotifer	Algae	−16**	Yoshida *et al.* (2007)
19	PP	Mite	Mite	-6^{ns}	Huffaker (1958)
20	PH	Parasitoid	Insect	8^{ns}	Utida (1957)
21	PH	Parasitoid	Insect	−22**	Utida (1957)

**$P \leq 0.05$; * $0.05 < P \leq 0.1$ (i.e., significant in a one-tailed test for EDI < 0); ^{ns} $P > 0.1$.

abundance coincide with minima in the resource species, and vice versa. Other data sets appear to start with a ¼-period lag and then become antiphase. One clear mechanism for antiphase consumer–resource cycling is rapid contemporary evolution of either prey defence or competitive ability as predator densities oscillate (Abrams & Matsuda 1997; Yoshida *et al.* 2003; Jones & Ellner 2007; Becks *et al.* 2012). When the predator is abundant, there is selection for effective prey defence, resulting in an increase in defended genotypes. This causes predator density to decline and then stay low even when prey are abundant. However, because defence has a cost, realised as poor competitive ability when prey are dense, there is selection for genotypes that are good competitors but consequently poorly defended. This in turn allows an increase in predator density. The result is antiphase evolutionary dynamics in which defended and competitive genotypes oscillate in frequency along with species abundance dynamics.

The theory of antiphase consumer–resource dynamics and the underlying eco-evolutionary mechanism has been known since Abrams & Matsuda (1997) documented them as one of several possible theoretical outcomes of contemporary rapid prey evolution. Yoshida *et al.* (2003, 2007) and Becks *et al.* (2012) demonstrated their occurrence for rotifer-algal (*Brachionus-Chlorella*/*Chlamydomonas*) microcosms in which antiphase cycles occurred in the presence of algal genetic variation for the defence-competitive ability trade-off, while cycles with a ¼-period lag occurred when prey genetic variation was absent. Moreover, identification of a specific trait underlying a defence-competition trade-off in the alga *Chlamydomonas* provided evidence for the expected accompanying oscillations of both heritable phenotypes (cell clumping) and gene expression (Becks *et al.* 2012). Finally, even when this interaction is initiated using algae without genetic variation, the system often transitions to antiphase dynamics after only two or three 'classic' predator–prey oscillations, presumably following the appearance and subsequent selection of a mutation (or mutations) producing defence. Algal populations are very large ($> 10^5$ cells/mL) and many generations occur before the transition to antiphase cycling, providing ample opportunity for mutation (generation times ~ 0.5 days, period of 'classic' cycles ~ 11 days, gives 20–30 generations before transition). Although the amount of genetic variation present and the shape of the fitness trade-off can lead to interesting variation in consumer–resource dynamics (Meyer *et al.* 2006; Yoshida *et al.* 2007; Becks *et al.* 2010), we focus here on the clearest indication of evolutionary predator–prey cycling: antiphase dynamics.

In exploring published studies of consumer–resource oscillations, we observed that the literature is replete with examples that exhibit antiphase dynamics for some or all of the period of observation. In addition to our rotifer-alga studies, these include phage attacking bacteria, ciliates and amoebae consuming bacteria, ciliates eating other ciliates, mites eating other mites and insects attacked by parasitoids. Furthermore, our experience that antiphase dynamics often appear in rotifer-algal microcosms after a few 'classic' cycles raised the question of whether these previous studies of consumer–resource dynamics also contained previously unrecognised effects of evolution. Could it be that undetected prey evolution is one factor in the scarcity of classic predator–prey cycles with a ¼-period lag (Ives 1998)? To evaluate

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this possibility, we surveyed the literature and found 21 suitable data sets, starting with Gause's 1934 report on *Didinium* consuming *Paramecium* and ending with the Yoshida *et al*. (2007) study of rotifers consuming algae. We developed statistical methods to probe data sets for a transition from 'classic' ¼-period lag consumer–resource dynamics to antiphase evolutionary dynamics, which if found would strongly suggest a response to natural selection. Finally, we evaluated the likelihood that these patterns were the product of prey evolution through a meta-analysis relating our metric of evolutionary effect to prey population size and the number of prey generations: larger populations and more generations should in principle provide greater opportunity for ecologically relevant mutations to occur. Our results suggest that in tightly coupled consumer–resource interactions, eco-evolutionary dynamics may be fairly common.

ALTERNATIVE MECHANISMS FOR ANTIPHASE CYCLING

Although we argue that antiphase predator–prey cycles are usually diagnostic of rapid prey evolution, we recognise that other processes can, in theory, produce antiphase predator–prey cycles. In Appendix S1 (Supporting Information) we explore in detail the mechanisms underlying antiphase cycles observed under some conditions in published models (de Roos & Persson 2003; and others cited in Appendix S1). In these models, cycles can arise when the predator has delayed maturation and predator–prey interactions cause discrete predator cohorts to form, whose nearly synchronous maturation and subsequent reproduction create a new cohort. Each cohort's maturation causes the predator population to start increasing, whereas the prey population starts decreasing because of predation by the new juveniles. Antiphase cycles (between cell number and nutrient concentration) also arise in a conceptually similar model for single-celled phytoplankton, in which progress through an early "juvenile" phase of the cell cycle depends on nutrient availability and cohorts form through cell-cycle synchronisation (Pascual & Caswell 1997). In these "cohort-driven" cycles, consumer minima coincide with resource maxima, but whether consumer maxima also coincide with resource minima (yielding an antiphase cycle) is just a matter of model details and parameter values: there is no specific timing relationship. In contrast, both classic ¼-period lag predator–prey cycles and ½-period lag eco-evolutionary cycles are characteristic consequences of underlying dynamic mechanisms. However, we explain in Appendix S1 that cohort-driven cycles have distinguishing properties of their cycle periods: (1) periods are on the order of one to two generations (Murdoch *et al*. 2003), up to four with long-lived adults; (2) In systems with constant background mortality (e.g., washout mortality in chemostats), the period must be short enough for one cohort's reproduction to compensate for background mortality during a complete cycle. Cycle amplitudes are limited by adult reproductive rate. In contrast, eco-evolutionary and classic predator–prey cycles can have much longer periods and arbitrary amplitude. We return to these differences after presenting our analysis of published data sets, and show that they rule out cohort-driven cycles in almost all cases.

A second alternative mechanism substitutes changes in prey community composition for changes in genetic structure within one prey species, with a trade-off among species between defence against predation and competitive ability. For this reason, we only analysed studies with one predator and one prey species: for the observed dynamics to have resulted from the presence of multiple prey species, there would have to have been repeated occurrences of undetected or unreported prey species. Whereas unacknowledged contamination is plausible in any one study, it is highly unlikely to have been a recurring problem for many different investigators in many different laboratories, and with contamination by species having the appropriate trade-off to produce antiphase cycles.

Finally, if genetically based trait dynamics can produce antiphase cycling, what about non-genetic prey traits? One obvious possibility is prey defences induced when predator density is high. Various predator-induced traits have been documented for phytoplankton and ciliates (Tollrian & Harvell 1999), as have non-heritable behavioural responses and maternal effects. Cortez (2011) explored induced plastic change theoretically and concluded that phenotypic plasticity in prey defence traits could not produce antiphase dynamics, and frequently decreased the lag between predator and prey. An exception naturally occurs when an induced trait persists for many generations after the inducing stimulus is removed. In these cases, the dynamics could behave as if the trait were heritable (and would appear as heritable using quantitative genetic methods). However, at least for induced traits, slow gain and loss over multiple generations has not typically been observed for the kinds of organisms in the studies analysed here (Tollrian & Harvell 1999). Similarly, epigenetic mechanisms that cause transgenerational change in trait expression (e.g., Anway *et al*. 2005; Nelson *et al*. 2013) could result in apparent eco-evolutionary dynamics. Likely epigenetic effects have been documented in *Chlamydomonas* as a defence against expansion of transposons (van Dijk *et al*. 2005). Although the number of generations that such translation controls can persist is as yet unknown, it would be fascinating if epigenetics were involved. We argue, however, that well-known effects of genetic differences among lineages is for now a more parsimonious, or at least a more general explanation, especially when considering the broad phylogenetic array of taxa in the studies analysed here (Table 1).

Given these considerations, we believe that antiphase consumer–resource dynamics with a period longer than two to four consumer generations, and especially with an amplitude greater than the reproductive potential of the consumers, are diagnostic of an eco-evolutionary feedback loop driven by prey evolution. We hasten to point out, however, that absence of antiphase dynamics does not mean that prey evolution is not occurring. For example, Yoshida *et al*. (2007) reported cryptic dynamics in which oscillations of total prey population were not detectable, although prey genotypes were oscillating within the population.

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MATERIALS AND METHODS

Selecting and extracting data from published studies

To identify studies of consumer–resource interactions with data suitable for our reanalysis, we searched Web of Science and Google Scholar with key words such as "consumer resource dynamics", "predator prey", "host parasite", "population & cycles". From the published studies containing data on consumer–resource dynamics, we selected only those that exhibited consumer–resource oscillations or damped oscillations, and in which there were at least two complete cycles. We further limited our search to two-species consumer–resource interactions with sufficient data points per cycle to quantify cycle period and phase lags. Even with these stringent criteria, we identified 21 suitable published data sets (Table 1; Table S2, Fig. S2). For each of these, initial transients were omitted from our analyses, setting a cut-off point just prior to the first peak in resource species abundance.

Evolutionary dynamics index

Evidence for a transition from a classic ¼-period lag to antiphase eco-evolutionary cycles can be seen in a phase-plane plot of predator against prey densities (Figs 1 and 2). The dynamics shift from a roughly oval limit cycle (produced by standard consumer–resource cycles without prey evolution) to a pattern approaching a line with negative slope (when dynamics are exactly antiphase). That is, the phase plane trajectory shows increasing eccentricity (the ratio between the orthogonal long and short axes of an ellipse). We characterised this change using the inverse of the eccentricity because it is bounded between 0 and 1 and is therefore more suitable for statistical analysis and more stable against imprecision in estimating the shorter axis. To quantify the changes in phase lag across a data set, we defined an Evolutionary Dynamics Index (EDI) by evaluating the inverse eccentricity of the phase-plane trajectories for a moving window over the data series. This yielded a sequence of values which decreases over time if a defended genotype appears and its increasing influence causes the cycles to approach antiphase dynamics.

To compute the EDI, phase-plane trajectories were first estimated by applying a spline smoother [function smooth.spline in R (www.R-project.org)] to log-transformed abundance data for each population (Appendices S4 and S5). The spline smoothing parameter was chosen by ordinary cross-validation with maximum degrees of freedom equal to ⅔ the number of data points. Smoothed trajectories for predator and prey were then constructed by evaluating the fitted spline at 100 equally spaced time points between the start and ending times of the data series. To provide equal scaling for the predator and prey populations, each smooth trajectory was scaled to have standard deviation 1. These were analysed in moving windows, each containing 33 points of the interpolated smoothed trajectories, or 50 interpolated points when that was necessary to have each window include at least one complete oscillation of predator and prey. For each window, a principal components analysis (PCA) was performed on the interpolated points in the phase plane (function princomp in R) using the normalised spline smooths of the data. Inverse eccentricity for a window is defined as the ratio of the smaller (shorter) to the larger (longer) of the principal components. For many data points distributed uniformly on the boundary of an ellipse, this procedure yields exactly the inverse of the ellipse's eccentricity. Applying this procedure to all windows gives a sequence of inverse eccentricity values. We fitted a linear regression of

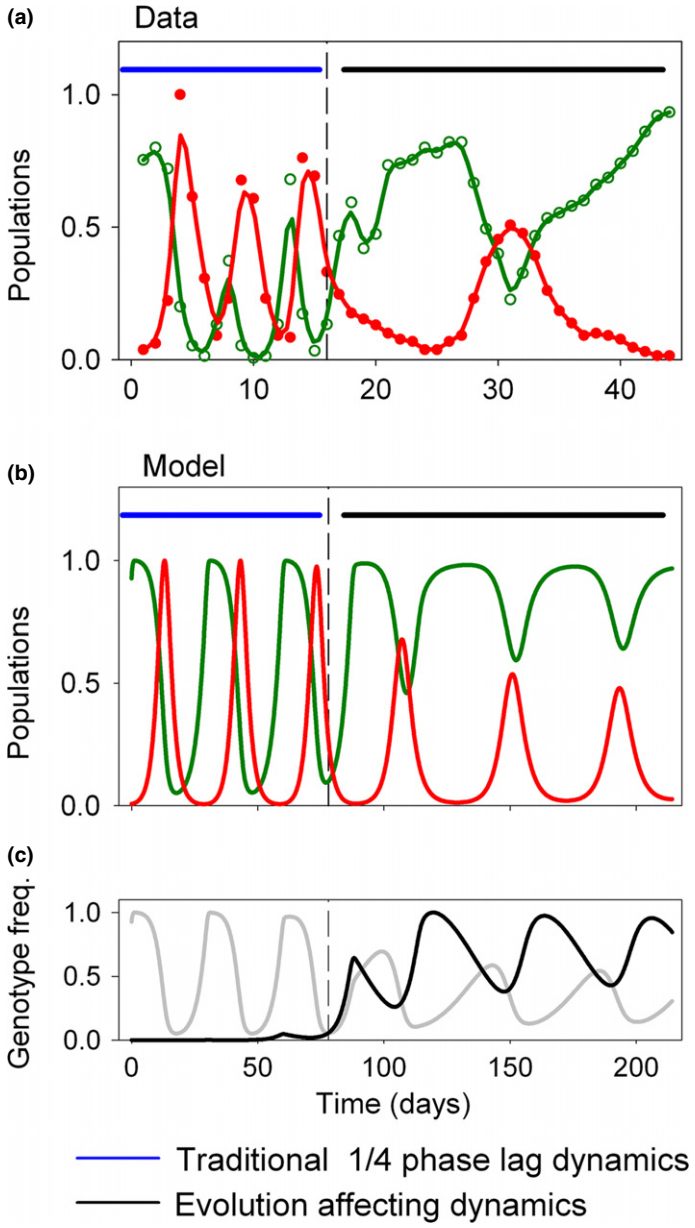


Figure 1 Transition from classic predator–prey cycles to antiphase cycles characteristic of eco-evolutionary dynamics. (a) Data for the amoeba, *Dictyostelium discoideum* (solid red circles) consuming the bacterium, *Escherichia coli* (open green circles), from Tsuchiya *et al.* (1972); data scaled to maximum abundance 1 for each species. (b) Population dynamics from an eco-evolutionary model in which defended prey are initially extremely rare. (c) Genotype dynamics for the model in (b) showing defended (black) and competitively superior (grey) genotype frequencies. Vertical dashed lines indicate the change point between classic and eco-evolutionary type dynamics (highlighted by blue and black lines at the top of panels a and b).

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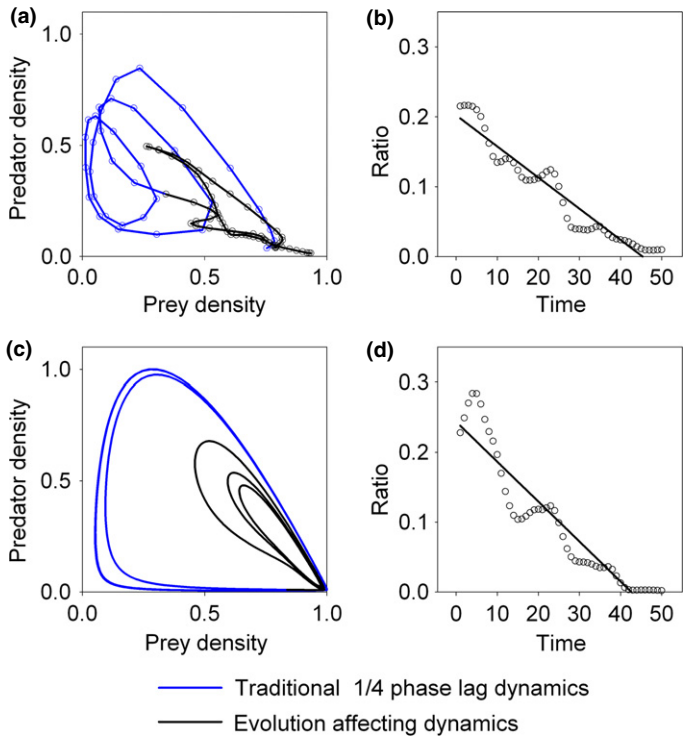


Figure 2 (a, c) Phase-plane plots from the amoeba-bacterium experimental data (Fig. 1a) and eco-evolutionary predator–prey model (Fig. 1b) respectively, with the initial interval of classic dynamics shown in blue and eco-evolutionary phase in black. (b, d) Eigenvalue ratios (circles) from principal components analysis of moving windows of the data in (a, c), showing the decreasing ratio between second and first eigenvalue. The solid line shows the linear regression of eigenvalue ratio on window number. The predicted change in eigenvalue ratio (linear regression slope × experiment duration) is our Evolutionary Dynamics Index (EDI), with a statistically significant negative value indicating a transition to antiphase dynamics (see Materials and Methods).

inverse eccentricity on window number, and defined EDI to be the difference between the first and last values of the fitted regression line. A statistically significant *negative* EDI indicates an *increase* in eccentricity, documenting a change from typical predator–prey oscillations towards antiphase cycles. Note that this analysis focuses on the change in eccentricity, not the degree of eccentricity *per se* at the start or end of the data series, or the slope of the cycles in the phase plane.

Significance of the EDI value was estimated by a residual bootstrap. The mean-square residual from smoothing the data is a biased estimate of measurement error variance. The bias was removed using a degrees-of-freedom correction based on the spline's effective degrees of freedom, multiplying all residuals by the appropriate factor such that their variance equalled the bias-corrected estimate of measurement error variance. For each data set, bias-corrected residuals were resampled and added to the fitted values (the spline evaluated at sampling times in the original data), and the procedure to compute EDI described in above was applied to the resulting bootstrap data set. This was repeated 5000 times for each empirical data set. Note that while the generation of replicates is based on a spline smooth, only the fitted values at the observation times (and no interpolated values) were used in generating the bootstrap replicates. EDI for a data set was assessed as significant at the 5% level if 95% or more of the resampled EDI values were less than zero. Because the inverse eccentricity values for some data sets exhibited a non-linear trend (e.g. a sigmoid decrease representing a rapid shift to antiphase cycles), we also tested significance using non-parametric correlation (Kendall's tau) between window number and inverse eccentricity. In this analysis, the EDI for a curve was assessed as significant (Fig. 3, Table S1) at the 5% level if 95% or more of the resampled tau values were less than zero. The results from this non-parametric test were nearly identical to the test based on linear regression, with one exception: Boraas (1980) was non-significant in the non-parametric correlation analysis.

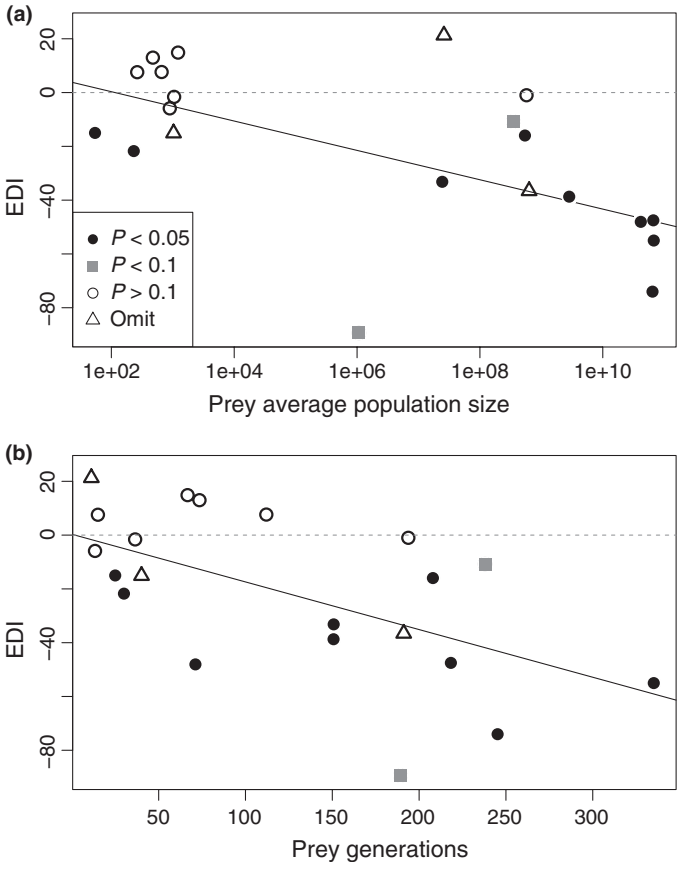


Figure 3 The development of antiphase evolutionary consumer–resource dynamics (evidenced by a negative value of Evolutionary Dynamics Index (EDI); see Fig. 2 legend and Materials and Methods) increases as a function of resource-species' (a) average population size and (b) number of generations, in the 21 published data sets in Table 1. Solid lines are linear regressions (a: $r^2 = 0.41$, $P < 0.01$; b: $r^2 = 0.31$, $P < 0.02$). Symbols indicate the two-sided statistical significance of individual EDI values: $P < 0.05$ (solid black circle), $P < 0.1$ (but significant at 5% level in one-sided test of EDI < 0; grey squares), $P > 0.1$ (open circles). Three studies (open triangles) were omitted from these statistical analyses; two because the cycles were already antiphase at or very soon after the start of the published data, the third because a negative EDI resulted from cycles becoming in-phase rather than antiphase (see Materials and Methods).

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Estimating number of prey generations

Most species in our data sets reproduce asexually by binary fission, and we inferred generation time from population doubling times. Two prey data series required different methods: for Utida (1957), we used generation times he reported, and for the prey in Huffaker (1958), *Eotetranychus sexmaculatus* generation times were estimated from data of Jamieson & Stevens (2007) on *E. sexmaculatus* temperature-dependent development, longevity and fecundity. The estimated number of prey generations was the duration of the data series divided by estimated prey generation time. Generation time could not be estimated for several predator species with different modes of reproduction. Supplementary Appendix S3 presents further details on these estimates.

Estimating average prey population size

In most cases, average prey population size was estimated from values in the original publications (e.g. Utida 1957; Huffaker 1958; Tsuchiya *et al.* 1972; Luckinbill 1973, 1974; Barnet *et al.* 1981; Bohannan & Lenski 1997; Yoshida *et al.*, 2003, 2007). For example, when data on both number of prey per volume and vessel volume were given, we calculated average prey population size directly. Where not all the necessary information was given (e.g., Gause 1934; van den Ende 1973; Jost *et al.* 1973 and Veilleux 1976), we had to make assumptions, e.g. concerning experimental vessel volume or conversion of biomass to population density. All of these necessary assumptions were based on previous studies conducted with the same or similar experimental systems.

Meta-analysis

We assessed the association between EDI and both population size and total number of prey generations to explore the likelihood that mutation followed by selection could explain significant EDI values. If prey evolution underlies antiphase cycling, mutations producing defence traits should be more likely in studies with larger prey population sizes, more prey generations, or both. Other mechanisms for antiphase cycles do not predict these patterns. To assess the association between EDI and population size and between EDI and number of prey generations, a permutation test was employed along with the residual bootstrap of EDI values. For each of 5000 bootstrap samples described above, EDI scores were permuted randomly across data sets and the Pearson product-moment correlation between the permuted bootstrapped EDI and each of the covariates was calculated. The *P*-value for each test was the proportion of the permutation correlations that was smaller (more negative) than the correlation between observed EDI values and the covariate. The same analyses were also performed using non-parametric correlation (Kendall's tau), with nearly identical results. In addition, we conducted an equivalent permutation test for the F-statistic of a linear regression of EDI on both prey generations and prey population size.

Three of the 21 data sets were omitted from this meta-analysis: two (Dulos & Marchand 1984 and Yoshida *et al.* 2003) because antiphase cycles were present from the start (our EDI analysis only works as a measure of the transition to antiphase dynamics if it occurs during a data series), and in the third data set (fig. 12a in Veilleux 1976) because the significant EDI value results from cycles becoming more in-phase rather than more out-of phase (based on the direction of the dominant PCA eigenvector). In-phase cycles generate a straight-line relationship in the phase plane, but with positive rather than negative slope.

RESULTS

A 1972 study by Tsuchiya *et al.*, on amoebae consuming a bacterium, initially led us to explore the potential role of evolution in a broader set of past studies. The similarity between the dynamics they observed (Fig. 1a) and those of a simulation model we wrote to describe rotifer–algal interactions (Fig. 1b; Yoshida *et al.* 2003; Ellner & Becks 2010) is striking, even though our model was built and parameterised some 30 years later, for different organisms. In our mechanistic model, prey genotypes along a trade-off between competitive ability and defence against predation coexist and develop long-term population dynamics. With the defended genotype initially very rare, predator and prey cycle with a classic ¼-period lag. During each peak in predator density, however, selection strongly favours the defended genotype, which eventually becomes common enough for the system to transition to antiphase eco-evolutionary cycles (Fig. 1b,c).

Although Tsuchiya *et al.* (1972) did not provide genotype-dynamics data, and indeed had no reason to collect those data, the correspondence between their observations and the predictions of our model is strongly suggestive of evolution of predator defence in the bacterium followed by eco-evolutionary antiphase cycling. Furthermore, our experiments where genotype or trait dynamics are actually known (Meyer *et al.* 2006; Becks *et al.* 2010, 2012) demonstrate that the presence of underlying evolutionary dynamics can be inferred from population abundance data: when cycle period is much longer than a single generation, antiphase consumer–resource dynamics is strong circumstantial evidence pointing to eco-evolutionary cycling. Consequently, when a shift from typical ¼-period lag to antiphase occurs, so that consumer–resource trajectories in the phase plane transition from a robust limit cycle to a pattern approaching a negative linear relationship (Fig. 2a,c), this is a pattern diagnostic of the *de novo* appearance or greatly elevated frequency of prey traits that confer defence against predation. It is this transition that we capture in our Evolutionary Dynamics Index (the change in inverse eccentricity of the phase-plane plot over time: see Materials and Methods). A statistically significant negative EDI value is taken to indicate an increase in the importance of evolutionary consumer–resource dynamics.

Of the 21 published studies of consumer–resource dynamics that we analysed, EDI was significant in 12 and marginally significant in one other, even though the original authors did not recognise (or have reason to recognise) that evolution might be acting, except for those in which we were

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involved (Table 1; Table S2; and excluding data in Fig. 12a of Veilleux 1976 for the reason explained above). Instead, the original authors either concluded that they had to some extent documented predator–prey cycling (Gause 1934), or expressed uncertainty about the underlying dynamics (Tsuchiya *et al*. 1972; Canale *et al*. 1973; van den Ende 1973; Jost *et al*. 1973; Barnet *et al*. 1981). Classic consumer–resource dynamics were never observed in Yoshida *et al*. (2003) because defended prey genotypes were present from the start (algae came from chemostats that had contained rotifers for months).

Finally, we found studies for which antiphase cycling was never detectable. This outcome was particularly prevalent in experiments with small prey population sizes and where dynamics were followed for relatively few generations, i.e. where mutations producing defence would be least likely to occur (Fig. 3). Indeed, the most substantial transitions to antiphase dynamics (strongest EDI) occurred in studies with large prey population sizes and many prey generations, where beneficial mutations for prey defence would be most likely to occur (EDI vs. population size, $r^2 = 0.41$, $P < 0.01$, Fig. 3a; EDI vs. generation number, $r^2 = 0.31$, $P < 0.01$, Fig 3B; EDI vs. generation number and population size, $F = 5.33$, $P < 0.02$: P values were calculated by a permutation test described in Materials and Methods).

EVALUATING AN ALTERNATIVE MECHANISM FOR ANTIPHASE CYCLING

At the end of the Introduction, we described possible alternative mechanisms for antiphase predator–prey cycling and explained why they are unlikely as explanations for the patterns analysed here. However, because our EDI analysis does not include direct evidence for prey evolution and its effects, we now return to consider the hypothesis of cohort-driven antiphase cycles. In eight of the 13 studies with significant or marginally significant EDI (Table 1), both predator and prey are single-celled organisms that reproduce by binary fission, allowing us to estimate generation times based on observed doubling times, as described above. Whereas the maximum period for cohort-driven cycles is about two generations, the observed periods are in every case much longer (Table S1.1 in Appendix S1) – in all but one case more than 10 times either predator or prey generation time (with the sole exception still being 4.0 prey and 7.8 predator generations). In the study where the "predator" is a virus, we cannot estimate virus generation time, but we show that the cycle period is much too long for the virus burst size to overcome the mortality due to washout from the chemostat during one complete cycle. In metazoans with long-lived adults, cohort-cycle periods can be up to four generation times. Utida's (1957) parasitoid–host data show cycle periods of six or seven generations, while for two of three data sets involving antiphase rotifer–algal dynamics we have shown experimentally that providing prey genetic variation transforms cycling from classic to eco-evolutionary antiphase dynamics (Yoshida *et al*. 2003, 2007). This leaves only the rotifer–algal system studied by Boraas (1980) for which we cannot rule out cohort-driven cycles in the predator. Finally, beyond the inability of cohort-driven dynamics to explain cycle period, it remains also unclear how cohort dynamics could explain the significant relationships between EDI and either prey generation time or prey population size, whereas both relationships are expected if the patterns result from the probability of a mutation for prey defence appearing and then being selected.

DISCUSSION

We found that evidence for eco-evolutionary dynamics is common in previously published data sets on two-species consumer–resource systems. Over half of the studies with sufficient data for our analysis support a conclusion that prey traits for defence against predation increased in frequency and altered the pattern of consumer–resource cycling. An additional study had antiphase eco-evolutionary cycles from the start of the published data series. In none of these studies, except those by Yoshida *et al*. (2003, 2007) which initially informed this analysis, did the original authors interpret their results in terms of contemporary eco-evolutionary dynamics. Our EDI analysis provides compelling evidence that evolution caused the quantitative changes in consumer–resource dynamics seen in many previously published time series.

In the cases presented here, the defended genotype (or genotypes) may have arisen by *de novo* mutation during the experiment and increased when consumers were abundant, or may have been present initially but too rare to influence the dynamics. In either case larger prey populations and more generations would make the appearance, or initial presence, of the essential variation and its developing effect on dynamics more likely, as we observed.

In most of these studies the defence trait is not known, indeed its presence was not suspected by the original authors. While many of the defences likely had to do with vulnerability to capture or ability to survive gut passage, it is not uncommon in microbial systems for some of the population to grow attached to the vessel. It might be speculated that "wall growth" could have been present in some studies and affected the dynamics. Wall growth would in fact produce antiphase cycling if there were distinct genotypes with and without the propensity for growing attached to the wall, if attached prey were less vulnerable to being consumed, and if there were a cost to having the trait. If there were no cost to wall growth, this trait would not produce antiphase dynamics because accumulation of attached prey would simply be a resource sink, leading to a decline of both predator and prey populations. In general, however, the nature of the defence should not affect the appearance of antiphase dynamics as long as the trait is genetically based. The pattern results from the rate of change in gene frequencies as determined by the effectiveness of the defence, its cost in competitive ability, and how strongly selection acts.

One conclusion from our analysis is that observing classic (purely ecological) consumer–resource cycles for longer than one or two oscillations may be difficult, because interference by rapid evolutionary change is likely: given sufficient time, adaptive evolution happens. This may underlie observations by some authors that good examples of classic consumer–resource cycles are difficult to find in the literature (Ives

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1998). At the same time, a pressing question in the study of eco-evolutionary dynamics is the extent to which they occur in natural populations, embedded in complex communities with many kinds of inter- and intraspecific interactions and residing in physically variable environments.

For some time it has been known that evolution can occur on the timescale of ecological interactions (e.g. Hendry & Kinnison 1999; Yoshida *et al*. 2003; Hairston *et al*. 2005). However, one can still ask how much ecologists are taking the possibility of evolution into account when they interpret their data. Our findings suggest that evolutionary dynamics are both common and often important in affecting outcomes, even in the relatively short period of laboratory ecological experiments. Many of the studies incorporated here are good examples of experiments that were originally planned to test purely ecological factors, but as we have shown, were clearly affected by evolution.

Our results contribute to a growing understanding that studies of community ecology will often benefit from observations at the trait and genotype levels in addition to traditional measures of species abundances in time and space. It is already appreciated that the individuals from a single species are not uniform units with identical properties: many species have multiple life history stages (Werner & Gilliam 1984; de Roos *et al*. 2003), more than one sex (Bierzychudek & Eckhart 1988), or inducible phenotypic responses to variable environments (Tollrian & Harvell 1999), each of which can affect dynamics. It is now also clear that genetically based trait differences, even within populations without age structure, sex or inducible phenotypes, can have dramatic effects on population interactions that must be understood if community ecology is to be a predictive science.

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AUTHOR CONTRIBUTIONS

TH designed research; TH, NGH, GH, LEJ and SPE performed research; GH and SPE contributed new analytic tools and analysed data; TH, NGH, GH, LEJ and SPE wrote the manuscript.

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